

Explanations for Decreased Harvest Index with Increased Yield in Soybean

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ABSTRACT

Yield in soybean [*Glycine max* (L.) Merr.] is controlled by total dry matter (TDM) accumulation and partition of that TDM into yield as measured by harvest index (HI). Previous research indicated that yield responded to TDM at R5 (seed initiation) [TDM(R5)] in an asymptotic relationship that plateaued at about 600 g m⁻². As yield rose with TDM(R5), HI declined, thus acting as a yield limiting factor. Explanation for the inverse relationship between yield and HI would help develop research strategies for yield improvement. Our objective was to determine if this negative relationship resulted from an inverse relationship between node production efficiency [node no. per gram TDM(R5)] with TDM(R5). Data for this study were collected from previous studies containing a variety of cultural treatments (planting dates, row spacings, plant populations, and waterlogging) conducted near Baton Rouge, LA (30° N lat), between 1987 and 1996 and combined into a single data set. Regression analyses between yield, TDM(R5), yield components, and production efficiencies of yield components [yield component level per gram TDM(R5)] were conducted. Results indicated that the inverse relationship between HI and yield ($R^2 = 0.54$; $b_1 = P < 0.0001$, $b_2 = P < 0.05$) resulted mainly from a negative relationship between node production efficiency and TDM(R5) ($R^2 = 0.81$; $b_1 = P < 0.0001$, $b_2 = P < 0.05$), which resulted in less efficient production of pods and seeds (the main yield components controlling yield) as TDM(R5) increased. In conclusion, reduced node production efficiency with increased TDM(R5) created the inverse relationship between HI and yield.

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Abbreviations: HI, harvest index; TDM, total dry matter.

YIELD IN SOYBEAN is a function of dry matter accumulation and the partition of this dry matter into the seed (i.e., harvest index [HI]) (Hay and Porter, 2006). Across changing environmental conditions, yield has been shown to be largely controlled by seed number per area (Egli, 1998), a yield component determined shortly after R5 (Board, and Tan, 1995; stages according to Fehr and Caviness, 1977). As shown in a companion publication (Board and Modali, 2005), total dry matter at R5 [TDM(R5)] is strongly linked with yield (Board and Modali, 2005) because this is when maximal vegetative TDM is formed (Egli and Leggett, 1973), an event closely concomitant with maximal production of nodes, pods, and seeds which set yield potential (Board and Modali, 2005). However, yield and TDM(R5) were not linked in a linear fashion but in an asymptotic relationship (Board and Modali, 2005). Yield increased steeply with TDM(R5) at levels <300 g m⁻², less steeply above this level, and then reached a plateau level at about 600 g m⁻². This relationship of TDM(R5) with yield was explained by an inverse relationship of HI with yield or TDM(R5). Harvest index acted as a “yield restricting factor” that retarded the beneficial effect of increased TDM(R5) on yield. Explanation of why yield and HI are inversely linked could help identify research strategies that result in greater yield at TDM(R5) levels near or above 600 g m⁻².

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In the companion paper, environmental–cultural effects on TDM(R5) influenced yield by affecting node number per area, reproductive node number per area (node bearing at least one pod having at least one seed), pod number per area, and seed number per area (Board and Modali, 2005). These four yield components, as well as yield itself, responded to increasing TDM(R5) in a negative exponential asymptotic fashion. This suggests that as environmental conditions improve, the increased TDM(R5) becomes less efficient at producing yield (i.e., reduced HI) and the four yield components affecting it [i.e., reduced yield component production efficiency: yield component level per gram TDM(R5)]. According to this evidence, the inverse relationship between HI and yield may result initially from reduced node production efficiency with increasing TDM(R5), which is reflected in decreased production efficiencies for reproductive node number per area, pod number per area, seed number per area, and yield.

An alternative explanation for the inverse relationship between yield and HI may be compensation between complementary yield components that jointly influence production of a subsequent yield component, or yield itself. Yield components can be classified sequentially (Board et al., 1999) into primary complementary yield components (seed number per area and seed size) affecting final yield; secondary complementary yield components (seed per pod and pod number per area) affecting seed number per area; tertiary complementary yield components (pod per reproductive node and reproductive node number per area) affecting pod number per area; and quaternary

complementary yield components (node number per area and percent reproductive nodes) affecting reproductive node number per area. Percent reproductive nodes is the percentage of all nodes that become reproductive.

Yield component compensation occurs whenever an increase in one yield component concomitantly results in a decrease in the complementary yield component, resulting in a constant level of the resultant yield component or yield itself. For example, reduced HI with increasing TDM(R5) may result from reduced seed size in response to greater seed number per area concomitant with the rising TDM(R5) (primary level); reduced seed per pod with increasing pod number per area (secondary level); reduced pod per reproductive node in response to rising reproductive node number per area (tertiary level); and/or reduced percentage reproductive nodes in response to rising node number per area (quaternary level). Although yield component compensation between seed number per area and seed size at the genetic level has been reported (Hartwig and Edwards, 1970; Board et al., 1999), occurrence on the environmental level has been inconsistent, with some reporting compensation between seed number and seed size (McAlister and Krober, 1958; Christy and Porter, 1982; Schonbeck et al., 1986), while others do not (Kollman et al., 1974; Vasilas et al., 1989; Board et al., 1999). Yield component compensation at secondary, tertiary, and quaternary levels has not been studied as well compared to the primary level, although negative correlations have been reported among pod per plant, seed per pod, and seed size in field beans (Duarte and Adams, 1972).

To gain a greater understanding of how dry matter accumulation is linked to HI and yield formation, our objective was to assess whether the inverse relationship between yield and HI was due to an inverse relationship between TDM(R5) and node production efficiency; and/or yield compensation between complementary yield components at the quaternary, tertiary, secondary, and/or primary levels.

MATERIALS AND METHODS

Data Set Construction

Data presented in the current paper are taken from the same six studies reported in a companion publication (Board and Modali, 2005). For reader convenience, descriptions of these studies (Board et al., 1990, 1992;

Table 1. A listing of journal articles from which data were obtained for the current study with various cultural, environmental, and genotypic data for individual studies conducted for soybean grown near Baton Rouge, LA, 1987–1996.

Reference	Years	Soil	Planting dates	Cultivars (MG)	Row spacing	Plant population	Irrigation
					cm	plants ha ⁻¹	
Board et al. (1990)	1987 1988	Mhoon silty clay [†]	22 May 1987	'Centennial' (VI)	100	220,000	No irrigation
			12 July 1987	'Forrest' (V)	50	325,000	
			24 May 1988				
			6 July 1988				
Board et al. (1992)	1989 1990	Mhoon silty clay	26 July 1989	Centennial (VI)	100	325,000	Irrigation applied
			10 July 1990		75		
					50		
					25		
Board and Harville (1996)	1992 1993	Commerce silt loam [‡]	15 July 1992	Centennial (VI)	100	222,000	No irrigation
			12 July 1993		50		
					25		
Carpenter and Board (1997)	1994 1995	Commerce silt loam	25 May 1994	'DP415' (V)	75	234,000	No irrigation
			23 May 1995			189,000	
						164,000	
						70,000	
Linkemer et al. (1998)	1993 1994	Commerce silt loam	12 July 1993	Centennial (VI)	100	220,000	No irrigation
			16 Aug. 1994		25		
Board (2000)	1995 1996	Commerce silt loam	23 May 1995	'DP3606' (VI)	75	390,000	No irrigation
			17 May 1996			145,000	
						80,000	

[†]Fine-silty, mixed, nonacid, thermic Typic Fluvaquent.

[‡]Fine-silty, mixed, nonacid, thermic Aeric Fluvaquent.

Board and Harville, 1996; Carpenter and Board, 1997; Linkemer et al., 1998; Board, 2000) are repeated in Table 1. In all studies, experimental designs were randomized complete block with four replications in split plot or split-split plot arrangements. Similar to the companion paper, data in the current study were analyzed across rather than within studies to obtain the widest range for achieving our objectives, a procedure followed by other researchers in analyzing agronomic relationships (Hay and Andrew, 1989; Loomis and Connor, 1992; Purcell and Specht, 2004). The purpose is to use data from several studies to identify significant relationships, rather than to find similar trends within experiments.

The six selected studies involved a wide range of environments across a 10-yr period near Baton Rouge, LA (30° N lat). Treatment factors included planting dates, row spacings, plant populations, and waterlogging stress but had limited genetic variability. As described in each reference pertaining to a particular study, these treatment factors created differences in light interception and crop growth rate, as well as developmental timing (resulting mainly from temperature and photoperiod differences) that provided a wide range of TDM(R5), yield, yield components, and yield component production efficiencies for achieving our goals. Within the six studies, care was taken to avoid and/or ameliorate any abiotic and/or biotic stresses other than treatment factors. Irrigation (Table 1) and recommended pesticides were used as necessary.

Data Obtained

Data used for the current study included yield, TDM(R5), HI, and the following yield components: seed size (g per 100 seed); seed per pod (no.); seed number per area (no. m⁻²); pod number per area (no. m⁻²); pod per reproductive node (no.); reproductive node number per area (no. m⁻²) (node bearing at least one pod having at least one seed); node number per area (no. m⁻²); and percent reproductive nodes (%) (percentage of all nodes that become reproductive). Each data point was the mean of four replications. Methods for determination of yield components, yield, HI, and TDM(R5) were used according to Board and Modali (2005). Yield components were classified into the following categories:

1. Primary yield components affecting yield: seed number per area and seed size.
2. Secondary yield components affecting seed number per area: seed per pod and pod number per area.
3. Tertiary yield components affecting pod number per area: pod per reproductive node and reproductive node number per area.
4. Quaternary yield components affecting reproductive node number per area: percentage reproductive nodes and node number per area.

Production efficiencies for yield and yield components were determined as follows: yield production efficiency: yield/TDM(R5) (g g⁻¹); seed production efficiency: seed number per area/TDM(R5) (no. g⁻¹); pod production efficiency: pod number per area/TDM(R5) (no. g⁻¹); reproductive node production efficiency: reproductive node number per area/TDM(R5) (no. g⁻¹); and node production efficiency: node number per area/TDM(R5) (no. g⁻¹).

Data Analyses

Regression analyses were conducted on data points that were means of four replications for treatment combinations within particular studies. For example, if a study involved planting dates, cultivars, and row spacings (such as Board et al., 1990), data points were the means of the four replications for all planting date–cultivar–row spacing treatment combinations. Relationships describing yield vs. TDM(R5) and yield vs. HI (Fig. 1), and yield vs. seed number (Fig. 2), were repeated from the companion publication for reader convenience. All other regression relationships in the study were conducted with SAS regression (PROC GLM; SAS Institute, Cary, NC) in which linear,

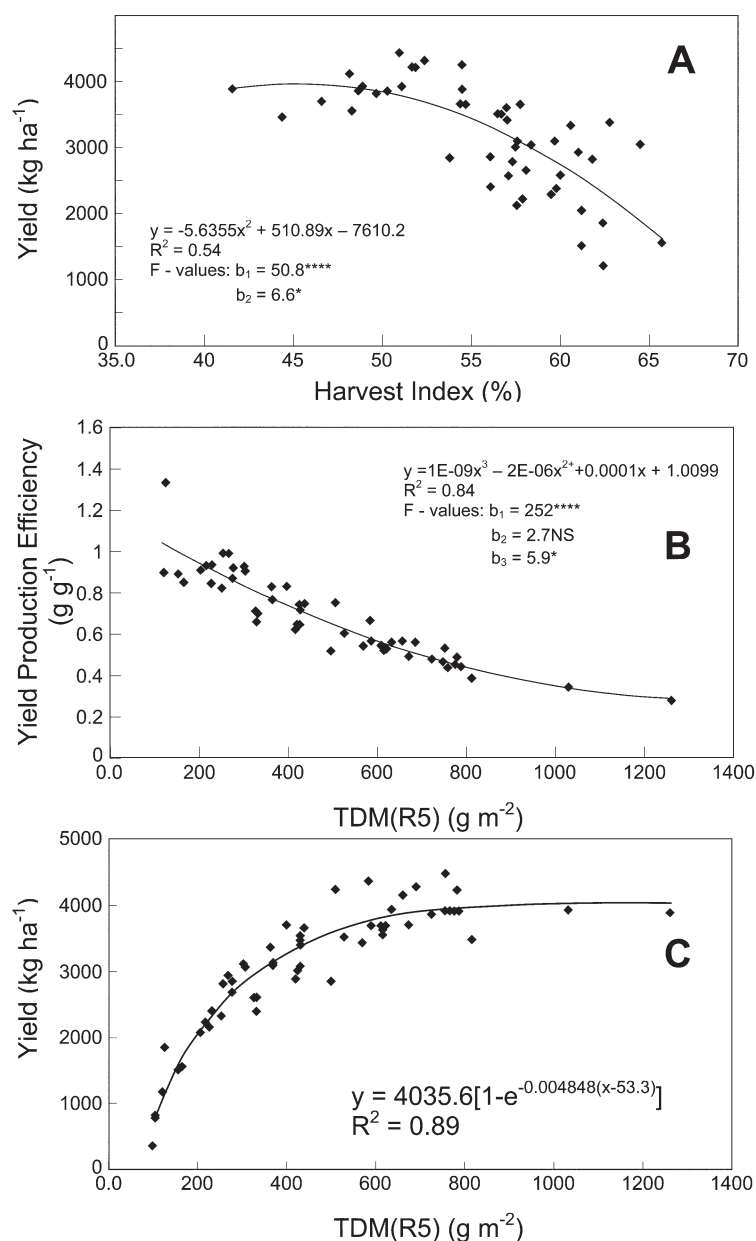


Figure 1. Relationships of (A) yield with harvest index, (B) yield production efficiency with total dry matter at R5 [TDM(R5)], and (C) yield with TDM(R5) for soybean grown across a range of environmental and cultural conditions near Baton Rouge, LA, 1987–1996. Each point is the mean of four replications for treatment combinations within specific studies making up the data set. *, **** Significant at 0.05 and 0.0001 probability levels, respectively. NS, not significant.

quadratic, and cubic components were successively tested for significance and included if the residual sum of squares was significantly reduced ($P < 0.05$). Identification of potential outliers was conducted using SAS PROC REG. Potential outliers were those having an R-student score > 2.0 . Two outlier points were identified in the 1994 waterlogging study. Because of extreme lateness of planting (16 August) for that year, TDM(R5) levels were very low ($100\text{--}125\text{ g m}^{-2}$) relative to data from the other

studies and therefore skewed results. Consequently, these two points were removed from the data analyses.

RESULTS

Yield Formation at the Primary Level

As demonstrated in the companion publication (Board and Modali, 2005), yield increased with TDM(R5) but did so in an asymptotic fashion ($R^2 = 0.89$, no P value available) (Fig. 1). Yield increased steeply with TDM(R5) at low dry matter levels ($< 300\text{ g m}^{-2}$), but these increases progressively declined as dry matter rose above this level. Yield reached a plateau level at a TDM(R5) of about 600 g m^{-2} (Fig. 1). This resulted in an inverse relationship between TDM(R5) and yield production efficiency ($R^2 = 0.84$; $b_1 = P < 0.0001$, $b_2 = \text{NS}$, $b_3 = P < 0.05$) (Fig. 1) within the 0 to 600 g m^{-2} TDM(R5) range. As TDM(R5) increased from 200 g m^{-2} to 800 g m^{-2} , yield production efficiency fell from 0.96 g of yield produced per gram TDM(R5) to about 0.32 g of yield per gram TDM(R5). Yield increased over this range, because the fourfold increase in TDM was greater than the 67% decrease in yield production efficiency. Reduced yield production efficiency with rising TDM(R5) resulted in an inverse quadratic relationship between yield and HI ($R^2 = 0.54$; $b_1 = P < 0.0001$, $b_2 = P < 0.05$) (Fig. 1). Thus, greater yields were associated with lower HI and lower yields with greater HI.

There was a strong linear relationship between yield and seed number per area ($R^2 = 0.83$; $b_1 = P < 0.0001$) (Fig. 2). In contrast, seed size played little role in explaining yield differences across the data set ($R^2 = 0.27$; $b_1 = P < 0.0001$). Seed production efficiency responded to TDM(R5) (Fig. 2) in a manner very similar to yield production efficiency (inversely quadratic; $R^2 = 0.81$; $b_1 = P < 0.0001$, $b_2 = P < 0.05$). At low TDM(R5) levels such as 200 g m^{-2} , about 8 seeds were produced per gram TDM(R5), whereas at greater TDM(R5) levels such as 800 g m^{-2} , only 3 seeds were produced per gram TDM(R5). Because no compensation was shown between seed size and seed number ($R^2 = 0.02$; $b_1 = \text{NS}$) (Fig. 2), the inverse relationship of yield production efficiency with TDM(R5) shown in Fig. 1 was entirely explained by the negative relationship of seed production efficiency with TDM(R5) shown in Fig. 2.

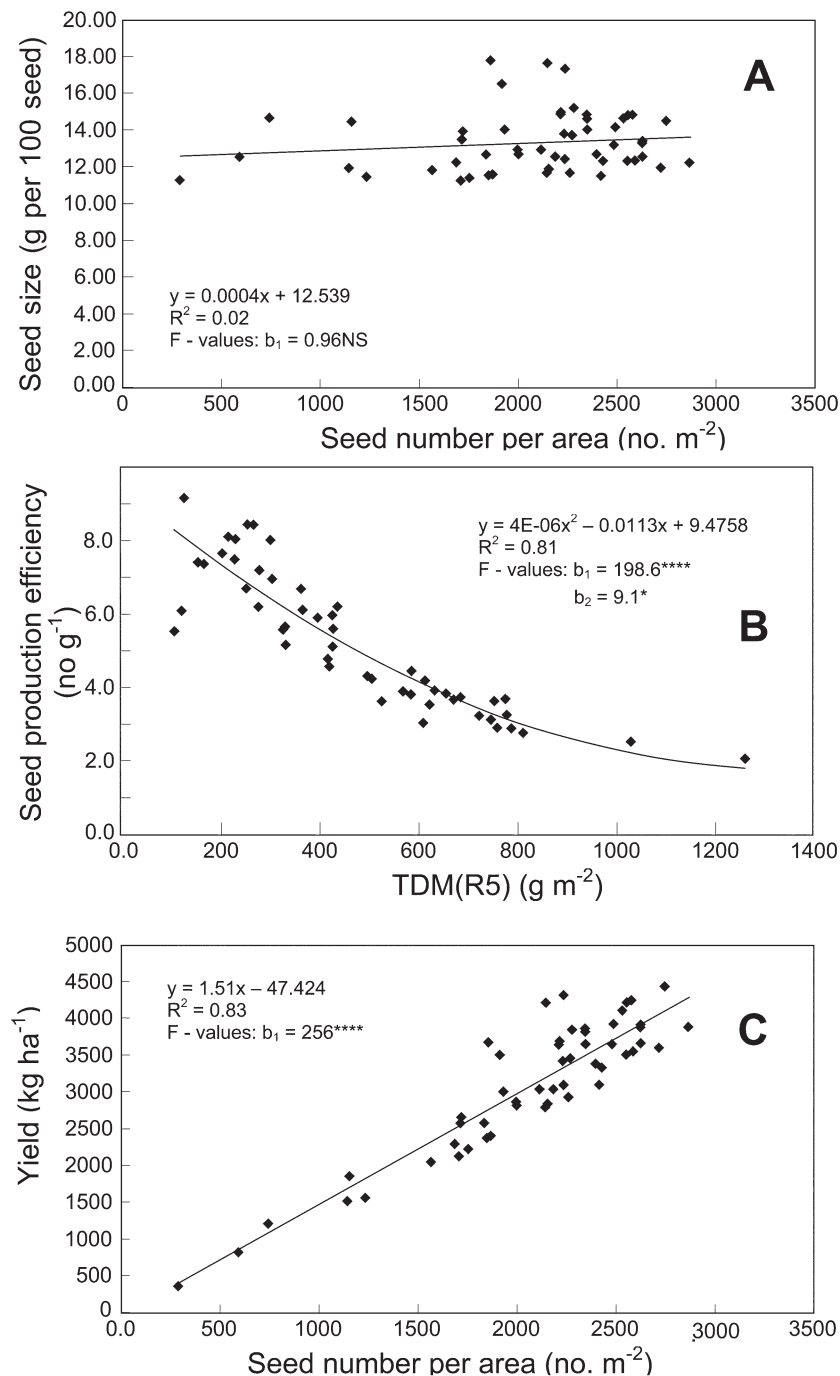


Figure 2. Relationships of (A) seed size with seed number per area, (B) seed production efficiency with total dry matter at R5 [TDM(R5)], and (C) yield with seed number per area for soybean grown across a range of environmental and cultural conditions near Baton Rouge, LA, 1987–1996. Each point is the mean of four replications for treatment combinations within specific studies making up the data set. *, **** Significant at 0.05 and 0.0001 probability levels, respectively. NS, not significant.

Yield Formation at the Secondary Level

Seed number per area was completely controlled by pod number per area (Fig. 3), with seed per pod having little influence ($R^2 = 0.03$; $b_1 = \text{NS}$). Seed number per area responded to pod number per area in a quadratic fashion ($R^2 = 0.91$; $b_1 = P < 0.0001$, $b_2 = P < 0.05$) but did not reach a plateau level. Pod production efficiency responded to increased TDM(R5) in a fashion similar to that shown for seed and yield production efficiencies ($R^2 = 0.83$; $b_1 = P < 0.0001$, $b_2 = \text{NS}$, $b_3 = P < 0.01$) (Fig. 3). Percentage drops in efficiency between high (800 g m⁻²) and low (200 g m⁻²) TDM(R5) levels were about 60% for pod number per area (Fig. 3), seed number per area (Fig. 2), and yield (Fig. 1). Although seed per pod and pod number per area were in a significant quadratic relationship (Fig. 3), the R^2 value was low ($R^2 = 0.32$; $b_1 = P < 0.0001$, $b_2 = P < 0.05$), as reflected in a slight 12% decline for seed per pod as pod number per area increased eightfold from 200 to 1600 pod m⁻². This lack of yield component compensation between seed per pod and pod number per area indicated that decreased seed production efficiency with increased TDM(R5) (Fig. 2) was predominately controlled by the inverse relationship between pod production efficiency with TDM(R5) (Fig. 3).

Yield Formation at the Tertiary and Quaternary Levels

Reproductive node number per area exercised greater influence over pod number per area ($R^2 = 0.74$; $b_1 = P < 0.0001$, $b_2 = 0.0001$) (Fig. 4) than did pod per reproductive node ($R^2 = 0.28$; $b_1 = 0.05$). Pod number per area responded to increased reproductive node number per area in a positive quadratic fashion ($R^2 = 0.74$; $b_1 = P < 0.0001$, $b_2 = P < 0.0001$) (Fig. 4). Pod number per area rose linearly as reproductive node per area increased to 400 node m⁻². Above this level, pod number per area was less responsive to increases in reproductive node per area and reached a plateau at about 600 reproductive node m⁻². Pod per reproductive node and reproductive node number per area did not demonstrate any kind of negative relationship ($R^2 = 0.26$; $b_1 = \text{NS}$, $b_2 = P < 0.0001$) (Fig. 4), thus indicating that yield component compensation was not occurring.

Reproductive node production efficiency displayed the similar inverse response to TDM(R5) ($R^2 = 0.79$; $b_1 = P < 0.0001$, $b_2 = P < 0.001$) (Fig. 4) shown by pod number per area, seed number per area, and yield. As

TDM(R5) increased from 200 to 800 g m⁻², reproductive node production efficiency fell 60% from 1.75 node g⁻¹ to 0.6 node g⁻¹. Because of the lack of yield component compensation between pod per reproductive node and reproductive node number per area, the inverse relationship of pod production efficiency to TDM(R5) shown in Fig. 3 resulted directly from the inverse relationship between

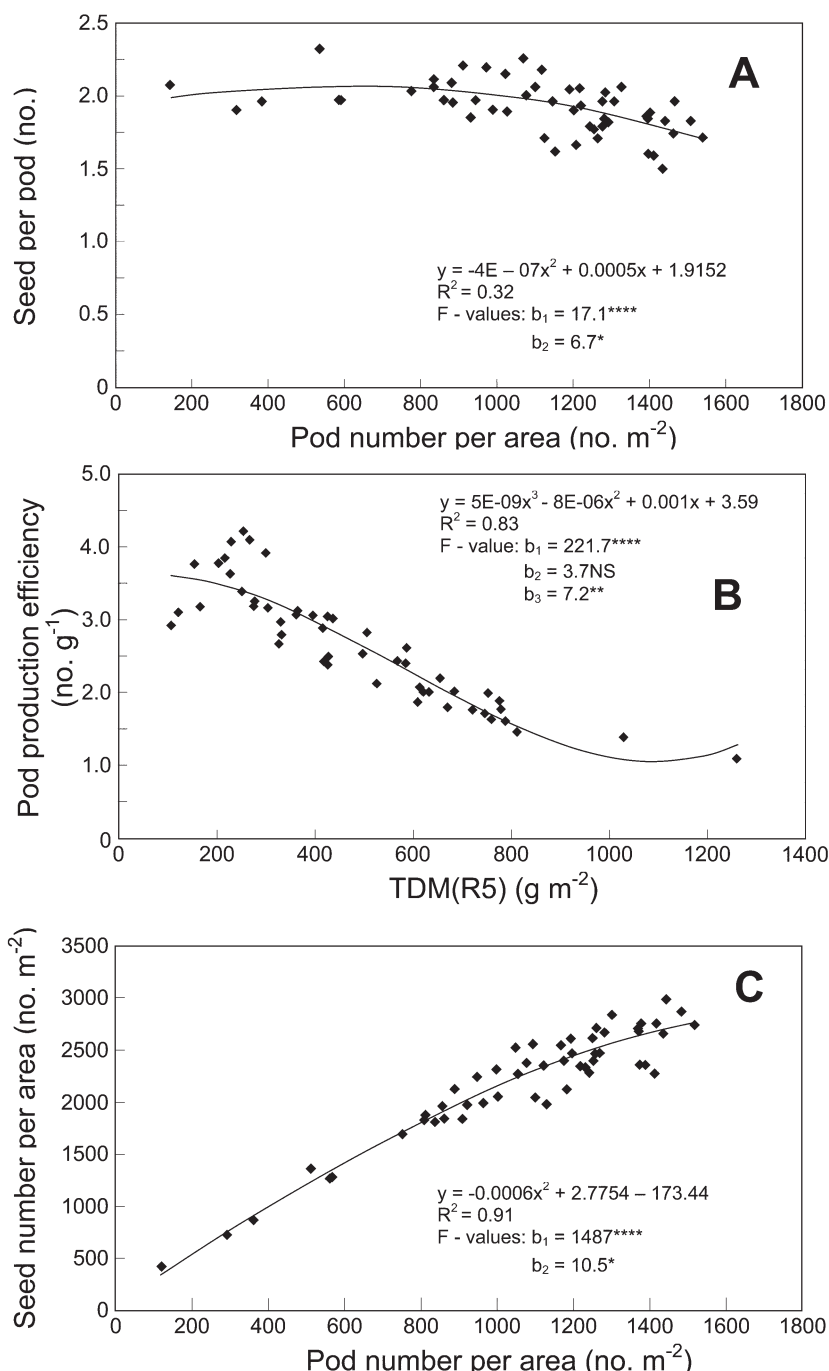


Figure 3. Relationships of (A) seed per pod with pod number per area, (B) pod production efficiency with total dry matter at R5 [TDM(R5)], and (C) seed number per area with pod number per area for soybean grown across a range of environmental and cultural conditions near Baton Rouge, LA, 1987–1996. Each point is the mean of four replications for treatment combinations within specific studies making up the data set. *, **, **** Significant at 0.05, 0.01, and 0.0001 probability levels, respectively. NS, not significant.

reproductive node production efficiency and TDM(R5) shown in Fig. 4.

Yield component responses at the quaternary level were similar to those on the primary, secondary, and tertiary levels. Reproductive node number per area and node number per area were closely linked in a linear

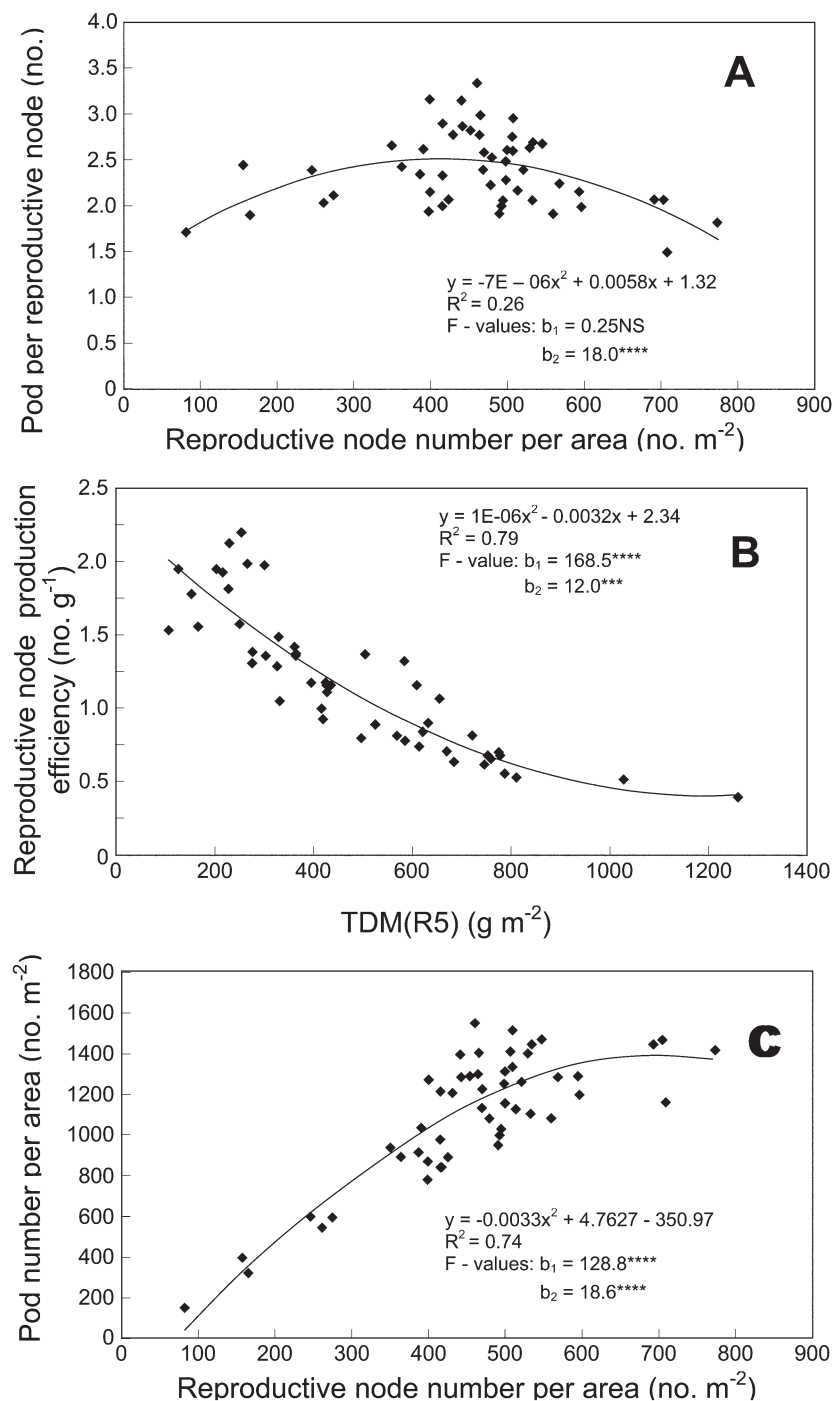


Figure 4. Relationships of (A) pod per reproductive node with reproductive node number per area, (B) reproductive node production efficiency with total dry matter at R5 [TDM(R5)], and (C) pod number per area with reproductive node number per area for soybean grown across a range of environmental and cultural conditions near Baton Rouge, LA, 1987–1996. Each point is the mean of four replications for treatment combinations within specific studies making up the data set. ***, **** Significant at 0.001 and 0.0001 probability levels, respectively. NS, not significant.

relationship ($R^2 = 0.94$; $b_1 = P < 0.0001$) (Fig. 5), whereas percentage reproductive nodes showed no influence on reproductive node number per area ($R^2 = 0.07$; $b_1 = NS$). Node production efficiency showed the same inverse quadratic relationship to increasing TDM(R5) ($R^2 = 0.81$; $b_1 = P < 0.0001$, $b_2 = P < 0.01$) (Fig. 5) as did reproductive node production efficiency (Fig. 4). At a low TDM(R5) level (200 g m⁻²), 2.3 nodes were produced for every g TDM(R5), whereas at a high TDM(R5) level (800 g m⁻²), only 0.65 nodes per gram TDM(R5) were produced. Lack of yield component compensation between node number per area and percentage reproductive nodes ($R^2 = 0.30$; $b_1 = NS$, $b_2 = NS$, $b_3 = P < 0.0001$) indicated that the inverse relationship between reproductive node number production efficiency and TDM(R5) in Fig. 4 resulted from the inverse response of node production efficiency to TDM(R5) (Fig. 5).

DISCUSSION

Results indicated that the inverse relationship between yield and HI, which limited yield responses to TDM(R5) as it rises (Fig. 1), clearly originated from the inverse relationship between node production efficiency with TDM(R5) (Fig. 5). This basic relationship then sets the pattern for how production efficiencies of other yield components important in affecting yield (reproductive node number [Fig. 4], pod number [Fig. 3], and seed number [Fig. 2]) respond to increasing TDM(R5). The end result is an inverse relationship between yield production efficiency and TDM(R5) (Fig. 1), which is reflected in the inverse relationship of yield with HI. This conclusion is supported by the consistency of relationships for production efficiencies of yield and yield components vs. TDM(R5) across the primary, secondary, tertiary, and quaternary levels. The hypothesis is also supported by the limited yield component compensation that occurred at all yield component levels, thus resulting in little confounding effect on the inverse relationship between yield component production efficiency vs. TDM(R5) as yield formation moved from the quaternary to primary levels.

Yield, yield components, and growth dynamic factors in the current study were largely modified through abiotic stresses (light interception and its effect on crop growth rate; and, in some cases, reduced developmental periods [created by photoperiod and temperature differences]) induced by the planting date, plant population,

row spacing, and waterlogging treatments. Similar results would have occurred with other abiotic stresses such as drought (Muchow et al., 1986), low soil pH (Mengel and Kamprath, 1978), compacted soil (Smucker and Allmaras, 1993), and mineral toxicities (Hanson and Kamprath, 1979) because these stresses also affect yield formation by reducing crop growth rate during the period between emergence and R5.

Lack of compensation between primary yield components in the current study agrees with some studies (Kollman et al., 1974; Vasilas et al., 1989; Board et al., 1999), but disagrees with others who reported compensation (McAlister and Krober, 1958; Christy and Porter, 1982; Schonbeck et al., 1986). Reports of increased seed size in response to reduced pod and seed numbers have occurred in cases where pod removal treatments were applied to selected plants that, up until the time of pod removal, had presumably been growing under optimal conditions (McAlister and Krober, 1958; Schonbeck et al., 1986). In the current study, changes in pod and seed numbers were created by differing cultural practices that altered environmental conditions during the growing season. Our data indicate that when suboptimal growing conditions between emergence and R5 result in low pod numbers, increased seed size cannot be expected to compensate for low pod numbers to maintain optimal yield (Fig. 2). Similar to the primary level, yield component compensation at the secondary (Fig. 3), tertiary (Fig. 4), and quaternary levels (Fig. 5) was not apparent. In summary, environmental differences that alter TDM(R5) predominately affect yield through node, pod, and seed production, with little confounding influence from percentage reproductive nodes, pod per reproductive node, seed per pod, or seed size.

Although the current study predominately deals with environmental influences on yield, results suggest avenues for genetic improvement. Greater yield through past cultivar improvement breeding programs may have resulted from increasing node number per area either through greater dry matter accumulation and/or greater node production efficiency. Among 18 modern and obsolete cultivars from the southern public cultivar pool, yield improvement was regulated by node, pod, and seed production (Board, unpublished data, 2007), a mechanism similar to that described in the current paper. However, greater node production resulted more from TDM accumulation

than greater node production efficiency. Although these observations must remain speculative until more definitive studies are done, observations suggest that within this group of cultivars, greater yield in modern vs. obsolete cultivars originated from greater node production.

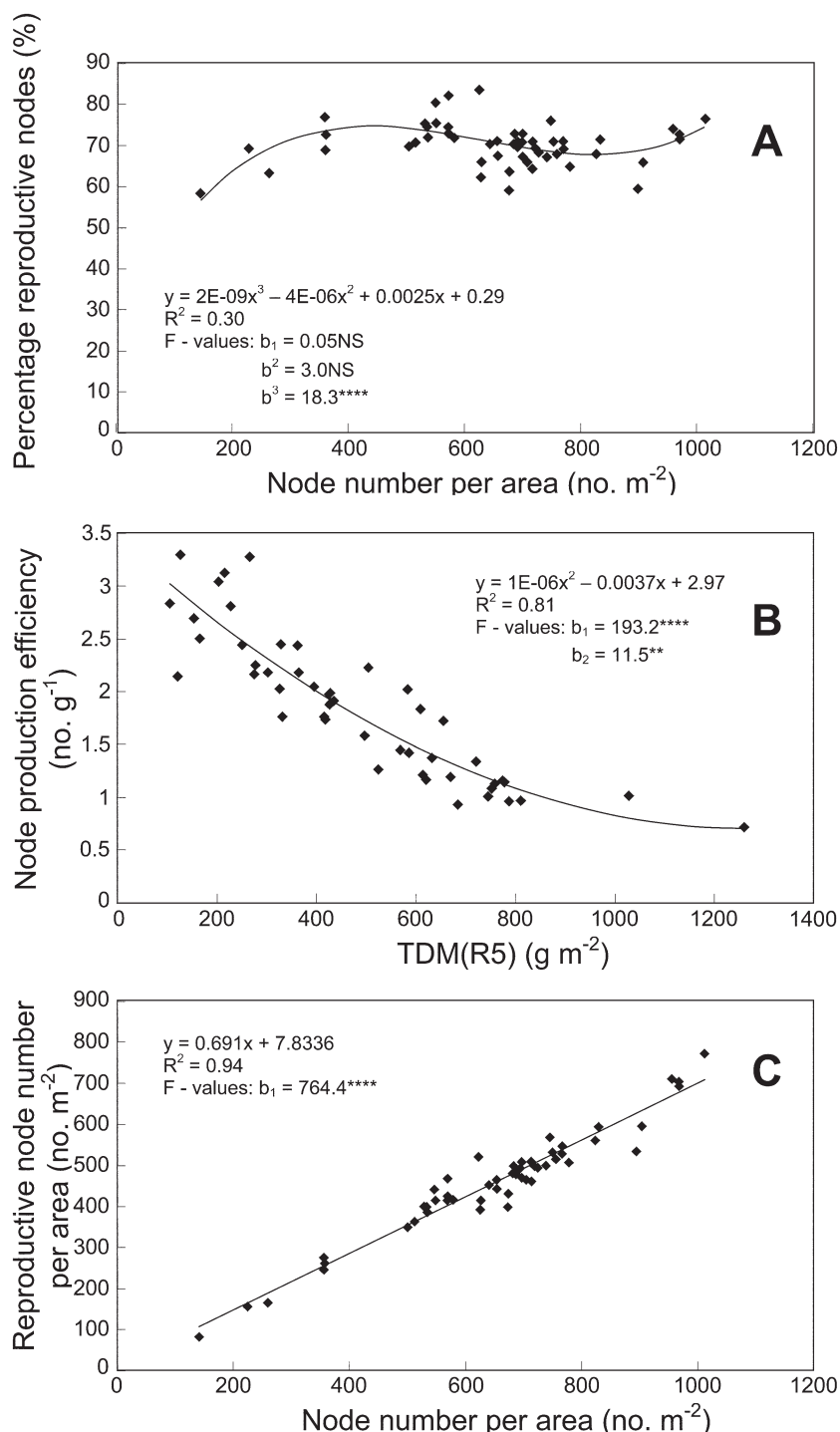


Figure 5. Relationships of (A) percent reproductive nodes with node number per area, (B) node production efficiency with total dry matter at R5 [TDM(R5)], and (C) reproductive node number per area with node number per area for soybean grown across a range of environmental and cultural conditions near Baton Rouge, LA, 1987–1996. Each point is the mean of four replications for treatment combinations within specific studies making up the data set. **, **** Significant at 0.01 and 0.0001 probability levels, respectively. NS, not significant.

CONCLUSIONS

The inverse relationship between HI and yield reported in the companion publication (Board and Modali, 2005) was shown to result from an inverse relationship between TDM(R5) with node production efficiency (Fig. 5). Because little yield component compensation occurred at any level of the yield formation process, the inverse relationship of TDM(R5) with node production efficiency was reflected in similar relationships of TDM(R5), with production efficiencies for reproductive nodes (Fig. 4), pods (Fig. 3), and seeds (Fig. 2), finally expressing itself in lower yield production efficiency as dry matter increased (Fig. 1). Thus, as yield rises with TDM(R5) to a plateau level, HI falls. Results support the view that environmental changes act on yield through control of TDM(R5) and the concomitant levels of nodes, pods, and seeds, and suggest that other yield components (fraction reproductive nodes, pod per reproductive node, seed per pod, and seed size) are more independent of environmentally induced changes on dry matter accumulation. Possible implications for these findings for genetic improvement of yield were discussed.

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